Manufacturing Operational Insights Report

Real Estate Triage Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: AAI-27

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Academic Year: Jan-Jun 2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features
2. Overall Architecture / Workflow
3. Tools & Technologies Used

Results & Output

Add the below details here:

1. *Screenshots / outputs*
2. *Reports / dashboards / models*
3. *Key outcomes*

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement and Objectives

1. Problem Statement

In real estate customer support systems, handling large volumes of incoming messages such as complaints, queries, and booking requests manually is time-consuming and inefficient. Delayed responses can lead to poor customer satisfaction and operational inefficiencies. There is a need for an intelligent system that can automatically analyze, classify, and respond to customer messages in real time.

2. Project Objectives

- To automate the triaging of real estate support messages
- To classify messages based on urgency and intent
- To extract important information using Named Entity Recognition (NER)
- To generate professional automated responses using LLMs
- To reduce response time and improve customer service efficiency

3. Scope of Project

- The system processes incoming text messages from users
- It supports classification, entity extraction, and response generation
- It generates structured reports for further processing
- It can be extended to real-world deployment with dashboards and analytics

Proposed Solution

1. Key Features

- Automated Message Processing – Incoming real estate support messages are analyzed automatically without manual intervention.
- Faster Issue Prioritization – Urgent complaints and critical requests are identified quickly for immediate action.
- Accurate Information Extraction – Important details such as names, dates, property IDs, and contact numbers are captured efficiently.
- Professional Response Generation – The system creates ready-to-use draft replies for customer communication.
- Improved Support Efficiency – Reduces manual workload and speeds up support operations.
- Structured Report Generation – Each request is converted into a clear JSON triage report for easy handling.
- Database Record Management – Reports are stored for future tracking, review, and analytics.
- Dashboard Monitoring – Admin users can view statistics, apply filters, and manage reports through the dashboard.
- Better Decision Making – Routing and SLA assignment help teams respond in a timely and organized manner.
- Scalable Full-Stack Solution – The project provides a strong foundation for future enterprise deployment and enhancements.

2. Overall Architecture / Workflow

The Real Estate Triage Agent follows a layered full-stack architecture consisting of frontend, backend, AI processing, and database modules.

The workflow begins when a user enters a support message through the web interface. The frontend sends this message to the Flask backend using a REST API request.

The backend receives the request, validates the input, and forwards the message to the AI processing pipeline. This pipeline uses a LangChain ReAct agent integrated with the Groq LLM.

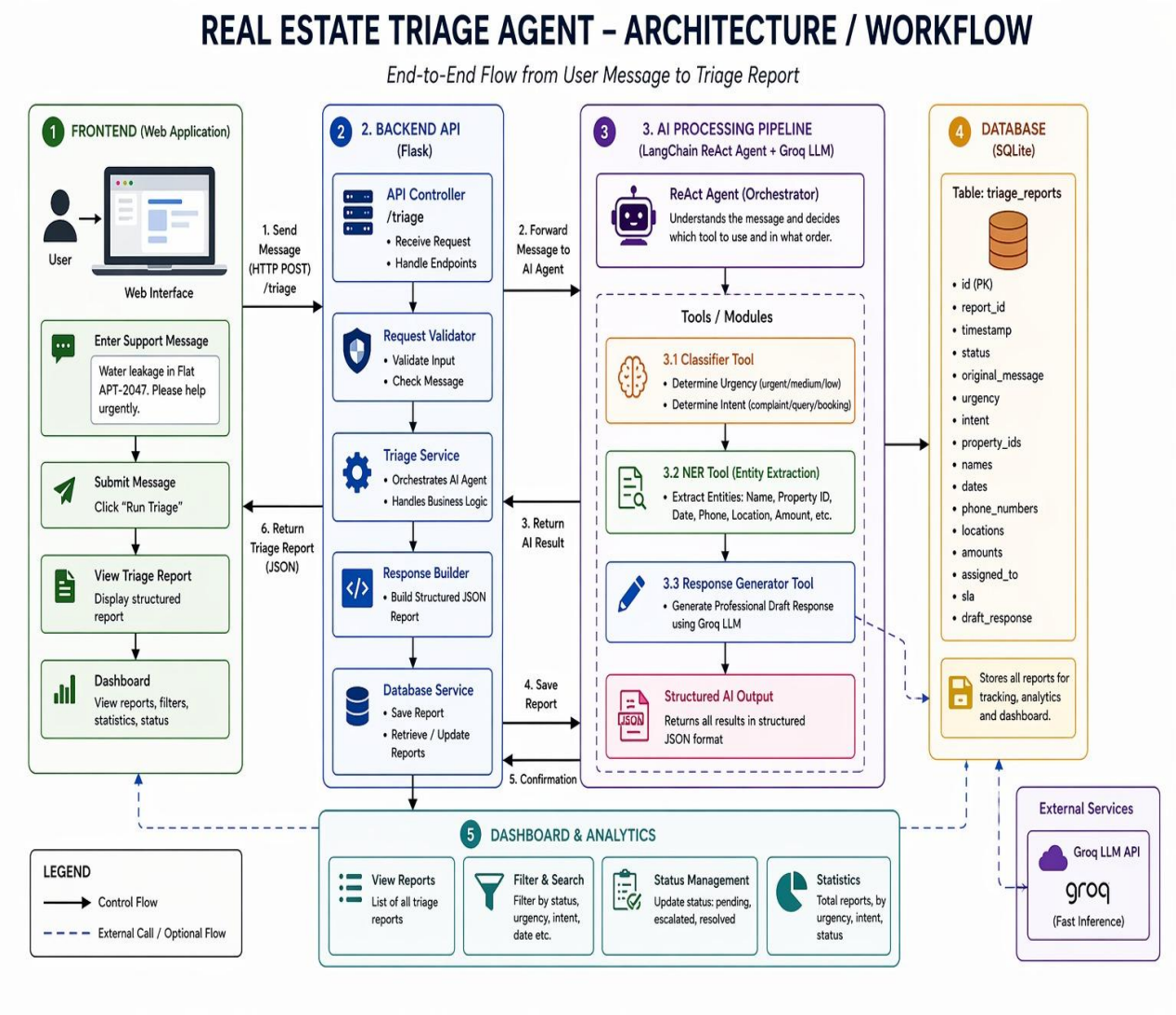
The agent coordinates three main tools:

- **Classifier Tool** → Determines urgency and intent
- **NER Tool** → Extracts entities such as names, property IDs, dates, and phone numbers
- **Response Generator Tool** → Generates a professional response

After processing, the system generates a structured JSON triage report. This report includes classification, extracted entities, routing information, and a draft response.

The report is stored in the database for future tracking and analytics. Finally, the processed output is returned to the frontend and displayed in a structured format.

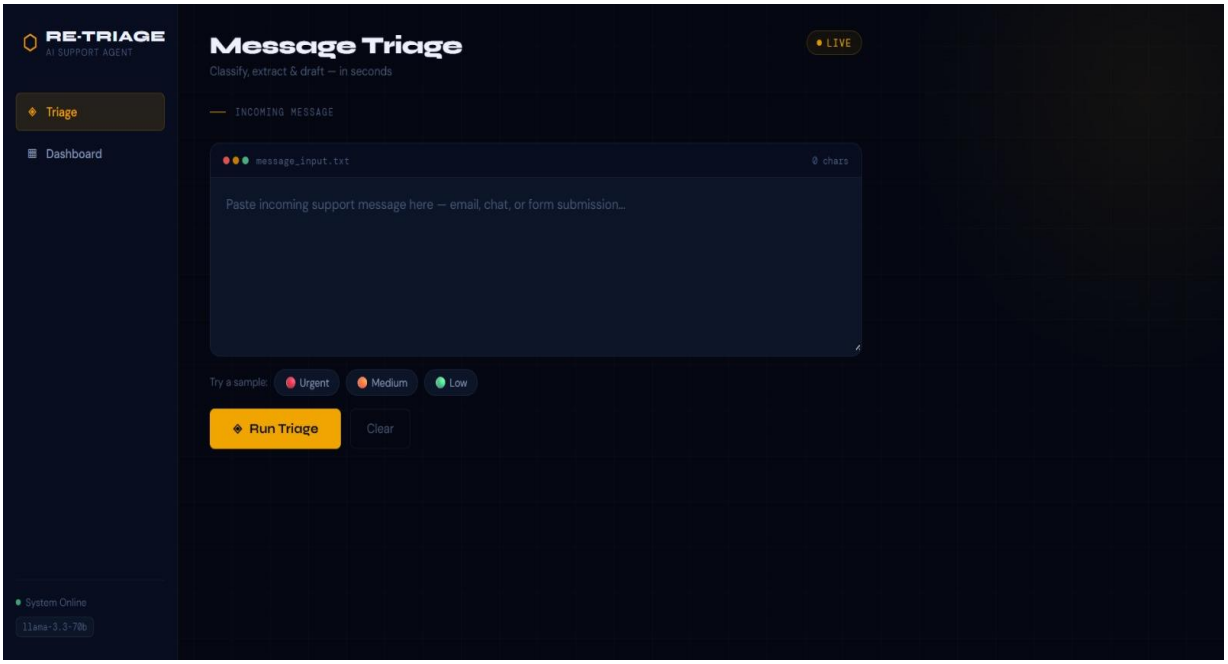
This architecture ensures automation, faster response handling, and efficient management of customer support operations.

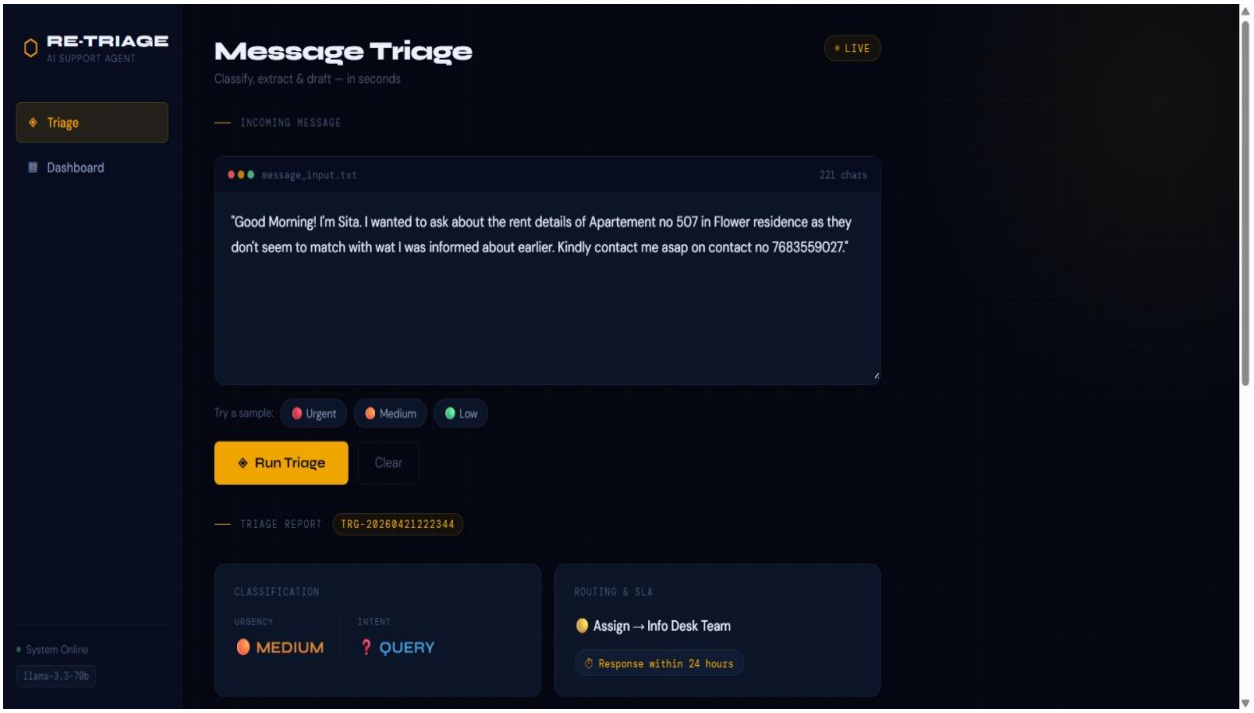


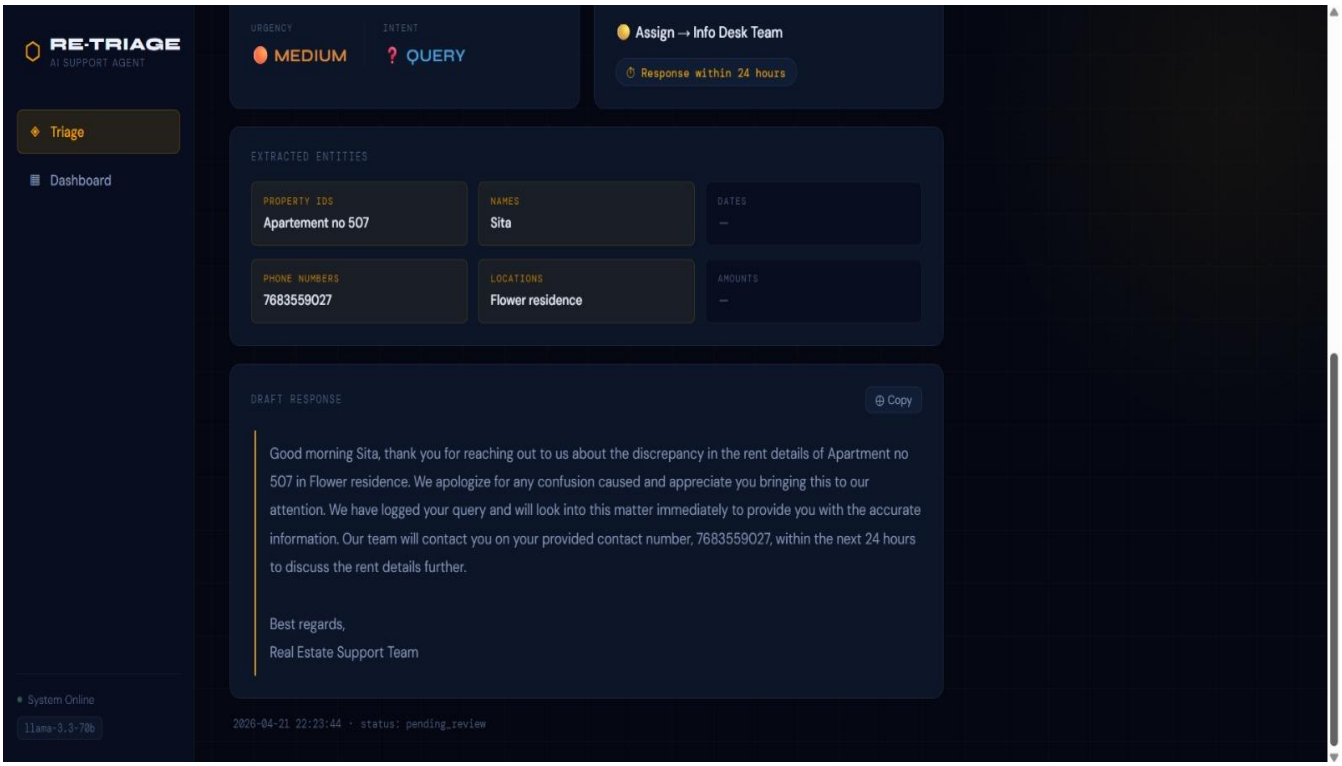
3. Tools and Technologies used

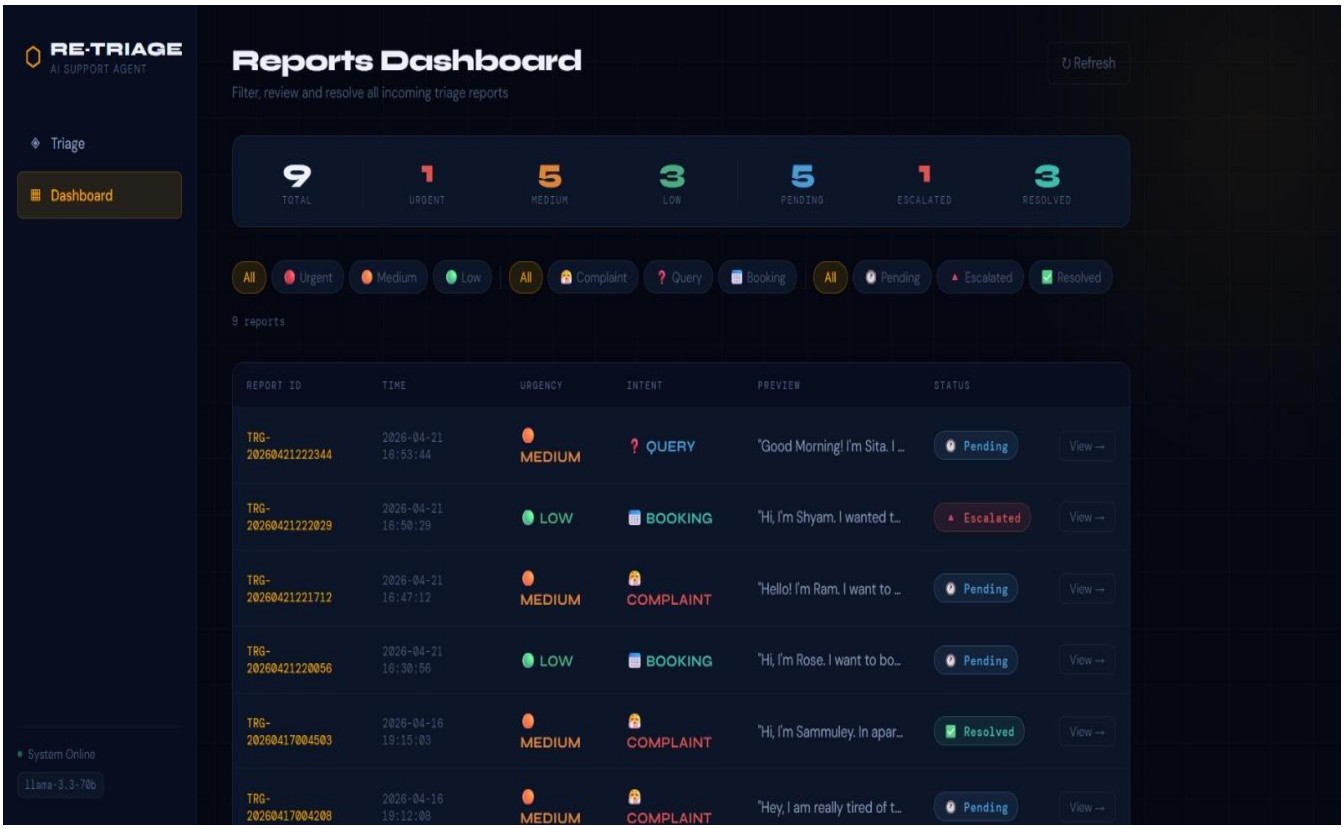
- **Programming Language:** Python
- **Framework:** Flask
- **AI Framework:** LangChain / ReAct Agent
- **LLM API:** Groq (LLaMA 3 Model)
- **Libraries:** spaCy, dotenv
- **Frontend:** HTML, CSS, JavaScript
- **Data Format:** JSON

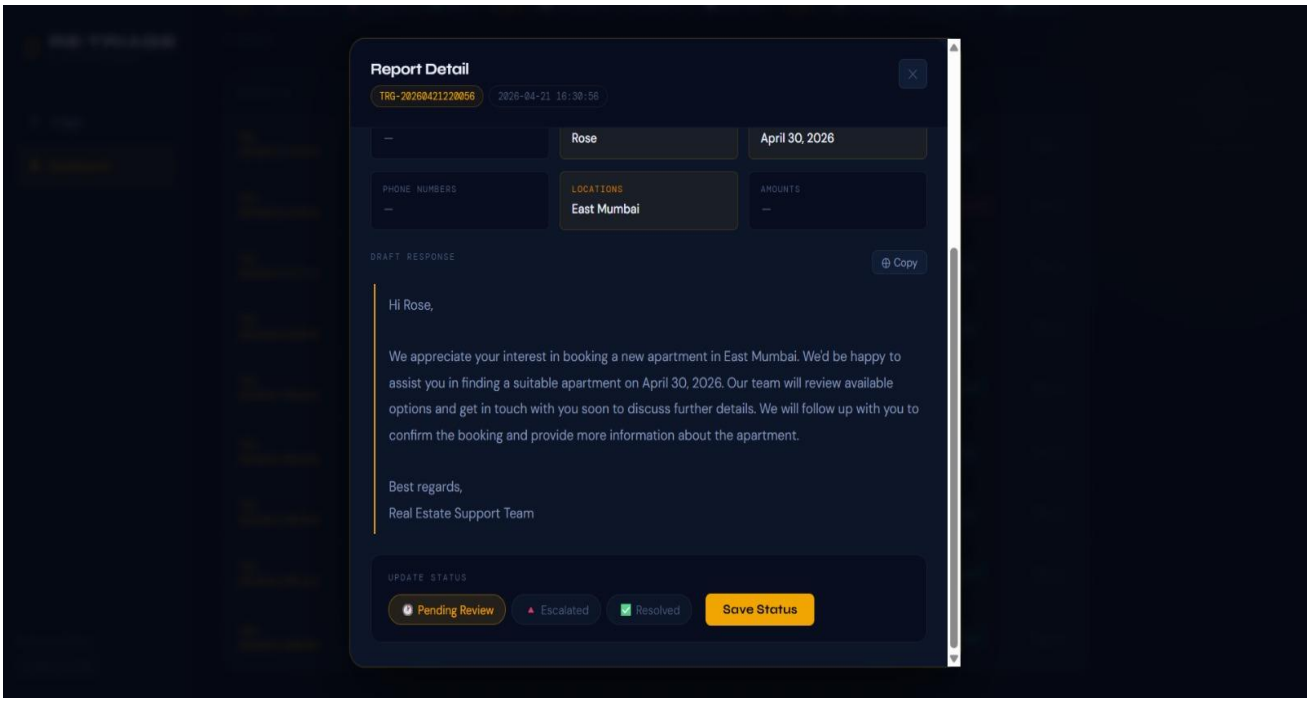
Results and Output






RE-TRIAGE
AI SUPPORT AGENT
Triage
Dashboard
System Online
llama-3.3-70b
URGENCY
MEDIUM
INTENT
QUERY
Assign → Info Desk Team
Response within 24 hours
EXTRACTED ENTITIES
PROPERTY IDS
Apartement no 507
NAMES
Sita
DATES
PHONE NUMBERS
7683559027
LOCATIONS
Flower residence
AMOUNTS
DRAFT RESPONSE
Copy
Good morning Sita, thank you for reaching out to us about the discrepancy in the rent details of Apartment no 507 in Flower residence. We apologize for any confusion caused and appreciate you bringing this to our attention. We have logged your query and will look into this matter immediately to provide you with the accurate information. Our team will contact you on your provided contact number, 7683559027, within the next 24 hours to discuss the rent details further.
Best regards,
Real Estate Support Team
2026-04-21 22:23:44 · status: pending_review


RE-TRIAGE
AI SUPPORT AGENT
Triage
Dashboard
System Online
llama-3.3-70b
Reports Dashboard
Filter, review and resolve all incoming triage reports
Refresh
9 TOTAL
1 URGENT
5 MEDIUM
3 LOW
5 PENDING
1 ESCALATED
3 RESOLVED
All
Urgent
Medium
Low
All
Complaint
Query
Booking
All
Pending
Escalated
Resolved
9 reports
REPORT ID
TIME
URGENCY
INTENT
PREVIEW
STATUS
TRG-20260421222344
2026-04-21 16:53:44
MEDIUM
QUERY
"Good Morning! I'm Sita. I ...
Pending
View →
TRG-20260421222029
2026-04-21 16:50:29
LOW
BOOKING
"Hi, I'm Shyam. I wanted t...
Escalated
View →
TRG-20260421221712
2026-04-21 16:47:12
MEDIUM
COMPLAINT
"Hello! I'm Ram. I want to ...
Pending
View →
TRG-20260421220056
2026-04-21 16:30:56
LOW
BOOKING
"Hi, I'm Rose. I want to bo...
Pending
View →
TRG-20260417004503
2026-04-16 19:15:03
MEDIUM
COMPLAINT
"Hi, I'm Sammuley. In apar...
Resolved
View →
TRG-20260417004208
2026-04-16 19:12:08
MEDIUM
COMPLAINT
"Hey, I am really tired of t...
Pending
View →


Report Detail
TRG-20260421220056
2026-04-21 16:30:56
Rose
April 30, 2026
PHONE NUMBERS
LOCATIONS
East Mumbai
AMOUNTS
DRAFT RESPONSE
Copy
Hi Rose,
We appreciate your interest in booking a new apartment in East Mumbai. We'd be happy to assist you in finding a suitable apartment on April 30, 2026. Our team will review available options and get in touch with you soon to discuss further details. We will follow up with you to confirm the booking and provide more information about the apartment.
Best regards,
Real Estate Support Team
UPDATE STATUS
Pending Review
Escalated
Resolved
Save Status

1. Key Outcomes

- Automated processing of customer messages
- Reduced manual effort in support handling
- Faster response time and improved efficiency
- Structured and actionable outputs for teams
- Scalable AI-based solution

Conclusion

The Real Estate Triage Agent successfully demonstrates the application of Agentic AI in automating customer support workflows. By integrating classification, entity extraction, and response generation into a single pipeline, the system significantly reduces manual effort and improves efficiency.

The use of a ReAct agent enhances decision-making by dynamically selecting tools based on the task. Overall, the project provides a scalable and intelligent solution for real-world customer support systems.

Future Scope and Enhancements

- Integration with real-time databases (MongoDB, Firebase)
- Deployment on cloud platforms (AWS, Azure)
- Adding dashboard analytics for insights
- Multi-language support
- Voice-based input processing
- Integration with CRM systems